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CHEMICAL REACTION ENGINEERING

FINAL EXAM

CHE 321, SPRING 2025, CLOSED BOOK, TIME: 60 MIN

The endothermic gas phase reaction $A \rightarrow B + C$ is carried out in a single steam-jacketed fluidized bed reactor, where the temperature of the steam circulating through the jacket remains constant at $T_a = 300°C$.

For the following data, calculate the steady-state reactor temperature.

Conversion: $X = 85\%$
Heat-transfer area of the jacket: $A = 5\ m^2$
Overall heat transfer coefficient of the jacket: $U = 500 \frac{J}{s.m^2.K}$
Heat of reaction per unit mole of species A at 25°C: $\Delta_r H = 120 \frac{kJ}{mol}$
Feed temperature: $T_0 = 40\ °C$
Feed molar flow rates: $F_{A0} = 10 \frac{mol}{s}$, $F_{I0} = 2 \frac{mol}{s}$ (inert), $F_{B0} = F_{C0} = 0$
Heat capacities (assume constant) $c_{pA} = 90 \frac{J}{mol.K}$, $c_{pB} = 60 \frac{J}{mol.K}$, $c_{pC} = 40 \frac{J}{mol.K}$, $c_{pI} = 50 \frac{J}{mol.K}$
Reactor energy balance: $\frac{dU}{dt} = \sum_\alpha F_{\alpha 0}\, h_\alpha(T_0) - \sum_\alpha F_\alpha\, h_\alpha(T) + \dot{Q} + \dot{W}_s$

Conversion $X \equiv \frac{F_{A0} - F_A}{F_{A0}}$

$\cancel{-(T-T_0)\sum_\alpha \Theta_\alpha C_{p\alpha} +}$

derivation

$$0 = +(-T+T_0)\sum_\alpha \Theta_\alpha C_{p\alpha} - X\Delta H^\circ_{rxn} - X\Delta C_p(T-T_R) + \frac{U}{F_{A_0}}A(T_a - T)$$

$$-T_0\sum_\alpha \Theta_\alpha C_{p\alpha} + X\Delta H^\circ_{rxn} - \Delta X C_p T_R - \frac{U}{F_{A_0}} A T_a = -T\sum_\alpha \Theta_\alpha C_{p\alpha} - TX\Delta C_p - T\frac{UA}{F_A}$$

$$\text{leftsterms} = -T\left(+\sum_\alpha \Theta_\alpha C_{p\alpha} + \Delta C_p X + \frac{U_A}{F_{A_0}}A\right)$$

$$T = \frac{+T_0\sum_\alpha \Theta_\alpha C_{p\alpha} \; X[-\Delta H^\circ_{rxn} + \Delta C_p T_R] + \frac{U}{F_{A_0}} A T_a}{+\sum_\alpha \Theta_\alpha C_{p\alpha} + \Delta C_p X + \frac{U_A}{F_{A_0}} A}$$

300C+273

$$T = \frac{0.85\left[-\left(180{,}000\,\frac{J}{mol}\right) + 10\frac{J}{mol\,K}(25C+273.15)\right] + (40+273.15)\left(100\,\frac{J}{mol\,K}\right) + \frac{500}{10}(5)\,T_a}{0.85(10) + 100 + \frac{500}{10}(5)}$$

$$T = \frac{75136.775}{358.5} = 209.6\,K$$

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r_A'

mole balance

$$F_{A0} - F_A + r_A' W = 0$$

$$F_{A0} X + r_A' W = 0 \rightarrow F_{A0} X = r_A' W$$

energy balance derivation

$$0 = \sum_\alpha F_{\alpha 0} h_{\alpha 0} - \sum_\alpha F_\alpha h_\alpha + \dot{Q} - W_s$$

$$F_\alpha = F_{A0}(\Theta_\alpha + \nu_\alpha X)$$

$$0 = \sum_\alpha F_{\alpha 0} h_{\alpha 0} - F_{A0} \sum_\alpha (\Theta_\alpha + \nu_\alpha X) h_\alpha + \dot{Q}$$

$$0 = \sum_\alpha \Theta_\alpha h_{\alpha 0} - \sum_\alpha \Theta_\alpha h_\alpha + \Delta H_{rxn} X + \frac{\dot{Q}}{F_{A0}}$$

$$0 = - \sum_\alpha \Theta_\alpha C_{p\alpha}(T - T_0) - \Delta H_{rxn} X + \frac{\dot{Q}}{F_{A0}}$$

$$\dot{Q} = -U(T - T_a)A.$$

$$T = \frac{X\left[\Delta H^\circ_{rxn} + \Delta C_p T_R\right] + T_0 \sum_\alpha \Theta_i C_{p\alpha} + \frac{U}{F_{A0}} A T_a}{X \Delta C_p + \sum_\alpha \Theta_\alpha C_{p\alpha} + \frac{U}{F_{A0}} A}$$

$$\Delta C_p = (40 + 60 - 90) \frac{J}{mol\,K} = 10 \frac{J}{mol\,K}$$

$$\sum_\alpha \Theta_i C_p = \Theta_A C_{pA} + \Theta_B C_{pB} + \Theta_C C_{pC} + \Theta_I C_{pI}$$

$$\Theta_A = 1, \quad \Theta_B = \Theta_C = 0, \quad \Theta_I = \frac{2}{10} = \frac{1}{5}$$

$$\sum_\alpha \Theta_i C_{pi} = 1(90) + \frac{1}{5}(50) = 100 \frac{J}{mol\,K}$$

T= 0.85